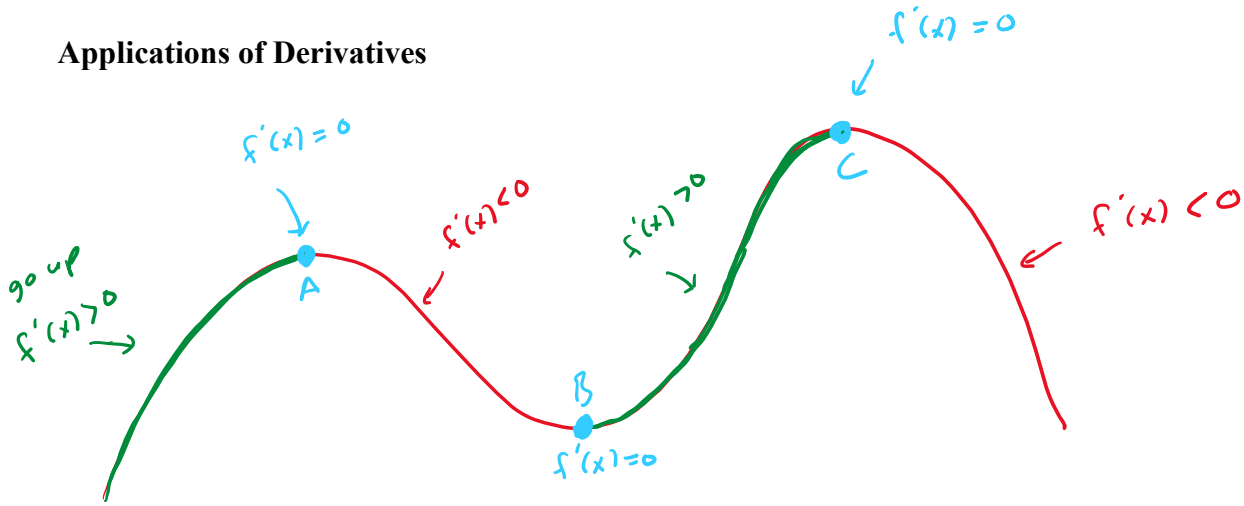
Applications of Derivatives



① A, B and C : Critical points

② A and C : local maxima

B : local minima

Finding the maximum or minimum of a function.

We can use the derivatives to find the minimum and maximum of functions. Suppose we want to find the minimum and maximum of a function $f(x)$. In these examples, we will be given that $f(x)$ will be maximized or minimized at the soltion of $f'(x) = 0$. Therefore, the procedure to do is as follows.

- Step 1: Find and simplify $f'(x)$
- Step 2: Solve $f'(x) = 0$. Suppose that the solution is $x = c$. (critical point, usually only 1 critical point in our examples)
- Step 3: Confirm that $\underbrace{f''(c)}_{\text{second derivative}} < 0$ for the maximum and $f''(c) > 0$ for the minimum.

1. **Cost Function, Marginal Cost Function**

   The marginal cost represents the approximate cost of producing the next unit.

Example 1:

The per-day cost function of the manufacture of portable MP3 players is given by

$C(x) = 128{,}000 + 30x + x^3$

where x is the number of MP3 players manufactured per day. Assume that the company cannot manufacture more than 100 MP3 players per day.

a. Find the $\underbrace{\text{marginal Cost}}_{MC}$ function.

$$MC = c'(x) = (128{,}000 + 30x + x^3)'$$
$$= (128{,}000)' + (30x)' + (x^3)'$$
$$= 0 + 30 + 3x^2$$
$$= 3x^2 + 30$$

b. $\underbrace{\text{Interpret the meaning of}}$ C' (80).

$$c'(80) = 3 \cdot (80)^2 + 30 = 19{,}230$$

$c'(80) \approx$ the cost of producing the 81st item.

c. Find the actual cost of producing the 81st item.

$C(81)$: cost of making 81 items

$C(80)$: cost of making 80 items

$$C(81) - C(80) = 661\,871 - 642\,400$$
$$= 19{,}471$$

d. Write the $\underbrace{\text{average cost}}_{AC}$ function

$$AC = \frac{\text{cost of making } x \text{ items}}{x}$$

$$AC = \frac{C(x)}{x} = \frac{128{,}000 + 30x + x^3}{x}$$

e. How many MP3 players should be manufactured in order to minimize the average cost?

Let $f(x) = AC$

$$f(x) = \frac{128{,}000 + 30x + x^3}{x}$$

Step 1: Find $f'(x)$

$$f(x) = \frac{128{,}000}{x} + \frac{30x}{x} + \frac{x^3}{x}$$

$$f(x) = 128{,}000 \cdot x^{-1} + 30 + x^2$$

$$\Rightarrow f'(x) = 128{,}000 \cdot (-1) \cdot x^{-1-1} + 0 + 2 \cdot x^{2-1}$$

$$\Rightarrow f'(x) = -128{,}000\, x^{-2} + 2x$$

$$f'(x) = \frac{-128{,}000}{x^2} + 2x$$

Step 2: Solve $f'(x) = 0$

$$\frac{-128,000}{x^2} + 2x = 0$$

$$2x = \frac{128,000}{x^2}$$

$$\Rightarrow \quad 2x \cdot x^2 = 128,000$$

$$\Rightarrow \quad 2x^3 = 128,000$$

$$x^3 = \frac{128,000}{2} = 64,000$$

$$x = \sqrt[3]{64,000} = 40$$

$$\boxed{x = 40}$$

Step 3: Verify that $f''(40) > 0$

$$f''(x) = (-128,000 \cdot x^{-2} + 2x)'$$

$$= -128,000 \cdot (-2) \cdot x^{-2-1} + 2$$

$$= 256,000 \cdot x^{-3} + 2$$

$$f''(x) = \frac{256,000}{x^3} + 2$$

$\Rightarrow \quad f''(40) = \dfrac{256,000}{40^3} + 2 > 0$

$\Rightarrow$ AC is minimized when $x = 40$.

<u>Note</u>: AC is minimized when AC = MC

You try

The per-day cost function of the manufacture of portable MP3 players is given by

$$C(x) = 686,000 + 30x + x^3,$$

where x is the number of MP3 players manufactured per day. Assume that the company cannot manufacture more than 100 MP3 players per day.

a. Find the marginal Cost function.
b. Interpret the meaning of C' (80).
c. Find the actual cost of producing the 81st item.
d. Write the average cost function
e. How many MP3 players should be manufactured in order to minimize the average cost?

2. **Revenue Function and Marginal Revenue Function**

It represents the approximate additional income gained from selling the next unit.

Example

A luxury watch manufacturer has determined that the daily price p (in dollars) they can charge is related to the quantity q they produce by the following quadratic demand function:

$$p(x) = -x^2 + 120x + 2700$$

a. Find the revenue function and the marginal revenue function

① $R = \text{price} * \text{quantity}$

$= (-x^2 + 120x + 2700) \cdot x$

$= -x^3 + 120x^2 + 2700x$

② $MR = R' = (-x^3 + 120x^2 + 2700x)'$

$= (-x^3)' + (120x^2)' + (2700x)'$

$= -3x^2 + 120 \cdot 2x + 2700$

$= -3x^2 + 240x + 2700$

b. Interpret the meaning of MR (10).

$MR(10) = (-3 \cdot 10^2 + 240 \cdot 10 + 2700)$

$= 4800$

c. Find the actual revenue of selling the 11st item.

$R(x) = -x^3 + 120x^2 + 2700x$

$R(11) - R(10)$

$= 42889 - 38000 = \boxed{4889}$

d. Find the quantity to maximize the revenue function

Step 1 : Find $R'(x)$

$R'(x) = -3x^2 + 240x + 2700$

Step 2 : Solve for $R'(x) = 0$

$-3x^2 + 240x + 2700 = 0$

$-3(x^2 - 80x - 900) = 0$

$\Leftrightarrow \quad x^2 - 80x - 900 = 0$

$\Leftrightarrow \quad (x - 90)(x + 10) = 0$

$x = 90$

$x = -10$

Only take positive solution : $x = 90$

Step 3 : verify that $R''(90) < 0$ for maximization

$$R''(x) = (-3x^2 + 240x + 2700)'$$

$$= -6x + 240$$

$$R''(90) = -6 * 90 + 240 = -300 < 0$$

$\Rightarrow$ Therefor $x = 90$ would maximize the revenue.

You try

A luxury watch manufacturer has determined that the daily price p (in dollars) they can charge is related to the quantity q they produce by the following quadratic demand function:

$$p(x) = -x^2 + 45x + 3000$$

a. Find the revenue function and the marginal revenue function
b. Interpret the meaning of MR (10).
c. Find the actual revenue of selling the 11st item.
d. Find the quantity to maximize the revenue function

3. Profit function and Marginal Profit Function

It represents the approximate change in total profit from producing and selling the next unit.

Example

A company determines that when x units of a product are produced each month, they will be sold at the price of

$$p = 22.2 - 1.2x$$

dollars per unit. The total cost of producing the x units will be

$$C(x) = 0.4x^2 + 3x + 40.$$

a. Find the profit function and the marginal profit function

Revenue = price * quantity

$\Rightarrow \quad R = p \cdot x = (22.2 - 1.2x)\, x$

$\Rightarrow \quad \boxed{R = 22.2x - 1.2x^2}$

Profit = Revenue − cost

$\Rightarrow \quad P = R - C$

$P = 22.2x - 1.2x^2 - (.4x^2 + 3x + 40)$

$P = 22.2x - 1.2x^2 - .4x^2 - 3x - 40$

$P = -1.6x^2 + 19.2x - 40$

$MP = P' = (-1.6x^2 + 19.2x - 40)'$

$$MP = -1.6 * 2x + 19.2$$

$$MP = -3.2x + 19.2$$

b. Interpret the meaning of MP(5).

$$MP(5) = -3.2 * 5 + 19.2 = 3.2$$

$MP(5) \approx$ The profit of selling the 6th item.

c. Find the actual profit of selling the 6th item.

$$P(6) - P(5) = 17.6 - 16 = 1.6$$

$$P = -1.6x^2 + 19.2x - 40$$

d. How many units should the company produce to maximize the profit?

$$P = -1.6x^2 + 19.2x - 40$$

Step 1 : $P'(x)$

$$P'(x) = MP = -3.2x + 19.2$$

Step 2 : Solve $P'(x) = 0$

$$-3.2x + 19.2 = 0 \iff x = \frac{19.2}{3.2} = 6$$

Step 3 : verify $P''(6) < 0$

$P''(x) = (-3.2x + 19.2)' = -3.2$

$\rightarrow P''(6) < 0$

Profit is maximized at $x = 6$

You try

. A company determines that when x units of a product are produced each month, they will be sold at the price of

$$p = 49 - x$$

dollars per unit. The total cost of producing the x units will be

$$C(x) = 0.125x^2 + 4x + 200.$$

a. Find the profit function and the marginal profit function
b. Interpret the meaning of MR(5).
c. Find the actual profit of selling the 6th item.
d. How many units should the company produce to maximize the profit?